

Amino acids nanocrystals for piezoelectric detection of ultra-low mechanical pressure

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ABSTRACT

Developing biocompatible nano-materials with the ability to detect ultra-low mechanical pressure is promising for biomedical sensors. This paper reports the detection of pressure as low as 1 Pa in the environmental pressure of 1 atm ($10^{-3}\%$ pressure change) by nanocrystals of amino acids glycine and alanine through the piezoelectric effect. Piezoelectricity enables detection of pressure by a change of dielectric polarization when the material is subjected to external pressure. This work exploits the non-centro-symmetric structure of some amino acids and their weak hydrogen bonds to develop sensitive mechanical pressure sensors. The β -glycine and L-alanine nanocrystals were grown from aqueous solution inside porous alumina substrate. The nanocrystals exhibit pronounced preferred crystallographic orientation. The sensitive piezoelectric response to ultra-low mechanical pressure is discussed based on atomic and crystal symmetry considerations.

1. Introduction

The ability to detect ultra-low fluctuations of mechanical pressure is highly desired in medical application devices [1–5], as well as in other technologies like gas leakage detection or artificial intelligence [1,6–8]. Following the significant progress that has been achieved during the past years in improving sensitivity of mechanical pressure detection by various methods, e.g. piezoresistivity, piezocapacitance and piezoelectricity [1], this paper proceeds one step forward. This research paper introduces amino acids as bio-materials for piezoelectric sensation of weak mechanical pressure changes of few Pascals in atmospheric ambient conditions, and shows its potential candidacy for biocompatible ultra-low pressure sensors.

Piezoelectricity is a linear coupling between dielectric polarization vector P_i ($i = 1-3$) and a strain tensor ϵ_j ($j = 1-6$) in non-centro-symmetric materials, through a tensor of piezoelectric coefficients d_{ij} , see Eq. (1).

$$P_i = d_{ij} \cdot \epsilon_j \quad (1)$$

Piezoelectric mode of mechanical pressure detection can be superior to other modes due to its low energy consumption and self-sustainability [9], in addition to its high sensitivity and short response time [10]. Recently, many studies showed improvement of the piezoelectric sensitivity to mechanical pressure changes in the range of kPa by utilizing thin films on flexible substrates [2,4,8,11,12]. Our recent work

lowered the mechanical pressure detection threshold to few Pa; it has been done by considering optimization of residual strains [13,14], low dielectric permittivity [15,16], growth with preferred crystallographic orientations [17] and weak interatomic chemical bonds. All these properties, in addition to biocompatibility, are satisfied in the amino acids nanocrystals. Here, we studied the simplest two among them, i.e. glycine and alanine [18,19].

Due to the need for piezoelectric biocompatible materials [1–5], a variety of bio-compatible piezoelectric nano-materials has been investigated, e.g. PVDF [1,20] and peptide micro- and nano-tubes [21–23]. The peptide nanotubes have been proven to have superior values of piezoelectric coefficients, whereas the PVDF is popular due to their processing as thin films and nanofibers despite its low piezoelectric coefficients. This paper introduces amino acids nanocrystals grown inside alumina nano-pores as promising bio-materials for piezoelectric detection of ultra-low pressure.

Glycine has three polymorphs in ambient conditions: thermodynamically stable γ -glycine, a metastable polymorph α -glycine and the non-stable β -glycine [24,25]. The three phases are distinguished by the angles of the chemical bonds between the glycine zwitterions within the crystal structure [25]. Glycine molecules in γ -phase form a three dimensional network of helical chains connected by hydrogen bonds [25–30]. The crystallographic structure of β and α -polymorphs is comprised of two dimensional sheet-like network of glycine zwitterions [25], which are connected by hydrogen bonds and form a monoclinic

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crystallographic unit cell [27,31]. The α -polymorph has a centrosymmetric unit cell that bans piezoelectricity, in contrast to the β -polymorph [32] which has a non-centro-symmetric structure with a polar axis in the b direction [31]. The piezoelectric tensor that refers to the crystal symmetry of β -glycine (space group $P2_1$) is [33]:

$$d_g = \begin{pmatrix} 0 & 0 & 0 & d_{14} & 0 & d_{16} \\ d_{21} & d_{22} & d_{23} & 0 & d_{25} & 0 \\ 0 & 0 & 0 & d_{34} & 0 & d_{36} \end{pmatrix} \quad (2)$$

Glycine crystals are usually grown by precipitation from liquid solutions; the polymorphism of glycine is controlled by modification of the parameters of the crystal growth process, e.g. acidity and compositions of the growth solution and the chemistry of the interface [24,27,30,34–41]. β -Glycine crystals can be stabilized inside nanoporous substrates due to a competition between thermodynamic and kinetic driving forces [42–45].

L-Alanine is one of the crystalline structures of the second basic amino acid alanine. L-Alanine has an orthorhombic unit cell (space group $P2_12_12_1$) that consists of a periodic alignment of zwitterionic alanine molecules connected by three dimensional asymmetric hydrogen bonds [46,47]. The polarization is created inside a unit cell when the material is elastically deformed by a shear stress, since only then the zwitterionic molecules align in a non-symmetric way through the distorted hydrogen bonds network. The crystal symmetry of L-alanine sets the piezoelectric coefficient to be as shown in Eq. (3) [33]. While the crystal growth of bulk L-alanine from an aqueous solution has been studied experimentally and theoretically [48,49], we could not find any reports on nanocrystals growth.

$$d_a = \begin{pmatrix} 0 & 0 & 0 & d_{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & d_{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & d_{36} \end{pmatrix} \quad (3)$$

This paper reports on crystal growth of β -glycine and L-alanine nanocrystals inside anodized aluminum oxide (AAO) porous substrate. The nanocrystals with preferred crystallographic orientation grow by precipitation from aqueous solution inside the nano-pores. The piezoelectric response of the nanocrystals of β -glycine and L-alanine to weak applied mechanical pressures (few Pa) is studied. The enhanced piezoelectric sensitivity within each of these materials is discussed based on their atomic structures.

2. Experimental

2.1. Fabrication of the nanocrystals

β -Glycine and L-alanine nano-crystals were grown inside alumina pores by a precipitation process from an oversaturated aqueous solution; this method is among the popular methods of crystal growth of functional nanocrystals [50–54]. The procedure of sample preparation is schematically described in Fig. 1. Prior to the crystal growth process, the porous substrates were prepared in the following way. Aluminum disks at a thickness of 0.1 mm and a diameter of 20 mm were mechanically polished to a roughness level of 10 nm. The AAO porous layer was obtained at room temperature by an electrochemical anodization of the polished surface in a 0.3 M oxalic acid aqueous solution under an applied electric field of 40 V/cm for 2 h. To enlarge the diameter of the alumina pores, the AAO porous layer was treated in a 5 wt % phosphoric acid aqueous solution by applying an electric field of 10 V/cm for 30 min.

To grow the β -glycine crystals, porous AAO substrates were placed in unsaturated aqueous solutions of β -glycine at different concentrations (1.8, 2 and 2.3 g/10 g H_2O) and held for 20 h in isothermal conditions of 24 °C. Variation of the conditions is highly important to control polymorphism. At the end of the growth process, the samples were taken out of the solutions and the outer surface of the porous layer

was delicately cleaned from any residual material. To grow L-alanine, a similar protocol was followed. The substrate was laid down in an aqueous solution of L-alanine (1.9 g/10 g H_2O) in 44 °C for 5 h, then taken out and cleaned from residual L-alanine on the surface.

Two electrical contacts were created to enable measurement of piezoelectric current generated by the nano-crystals inside pores. Measurement is carried out in parallel to the axes of nano-pores, i.e. in perpendicular to surface with one contact on the porous surface and another contact at bottom of the pores. The upper contact consists of a 1 mm diameter silver paste connected to a copper wire. The back side electrical contact is actually the aluminum substrate, connected to a copper wire by a silver paste. Hence, the electrical measurements correspond to electrodes on opposite sides of the longitudinal axes of pores. The surface geometry and thickness of the porous AAO layers were characterized using a HRSEM (Zeiss Ultra-plus FEG). The crystallographic phase and orientation of the grown β -glycine and L-alanine crystals were identified by performing XRD scans in the Bragg-Brentano configuration (Rigaku Smart Lab).

2.2. Experimental setup for the piezoelectric investigation

In order to study the sensitive piezoelectric response of the samples, an experimental system (Fig. 2) was designed to exert periodic mechanical pressures of 1–40 Pa and enable simultaneous measurements of the electrical current response. The porous surface of the samples was exposed to an air flow directed perpendicularly to the surface. The air flow was emitted from a 0.5 mm diameter needle-opening at a distance of 1 cm from the sample; its intensity was controlled by a mass-flow controller. The value of the mechanical pressure applied on the samples was measured for different valve openings of the mass flow controller and calibrated using a calibrated weight balance device (sensitivity of 50 μ g, OHAUS). The air flow was time-modulated by chopping the air flow with a rotating wheel with round holes, creating a periodic mechanical pressure profile at a frequency of 0.2 Hz. The area of incident airflow on samples was measured with circular aperture and found to be 5 mm. Cu–Sn wires were attached with a silver paste to the front and back surfaces of the samples and connected to a sensitive electric current-meter (Keithley 6154 System Electrometer). The piezoelectric current response was transferred from the current meter to an oscilloscope screen.

3. Results

3.1. Porous substrates

HRSEM top view images in Fig. 3a show a high areal density of pores (about 10^{11} pores/cm²) with an average diameter of 55 nm and a narrow diameter distribution. Fig. 3b shows a vertical cross section HRSEM image of the porous AAO layer; the pores are aligned vertically to the outer surface plane of the aluminum substrate. The depth of the pores is about 20 μ m. The walls of the pores, about 40 nm thick, separate neighboring pores and surround the bottom of the pores with a small radius of curvature.

3.2. Glycine nanocrystals growth

Since glycine is a polymorphic material, it is highly important to control the crystal growth process in order to obtain the desired piezoelectric phase, i.e. the less stable and symmetric polymorph. Control over the resulting phase of glycine was achieved by altering the concentration of the liquid solutions. Fig. 4 presents the XRD spectra of the specimen that correlates between the solution concentration and the obtained crystallographic glycine polymorph inside the nano-pores. X-ray diffractograms also show that the glycine crystals grow inside the pores with preferred crystallographic orientation where the (010) planes are parallel to the porous layer plane.

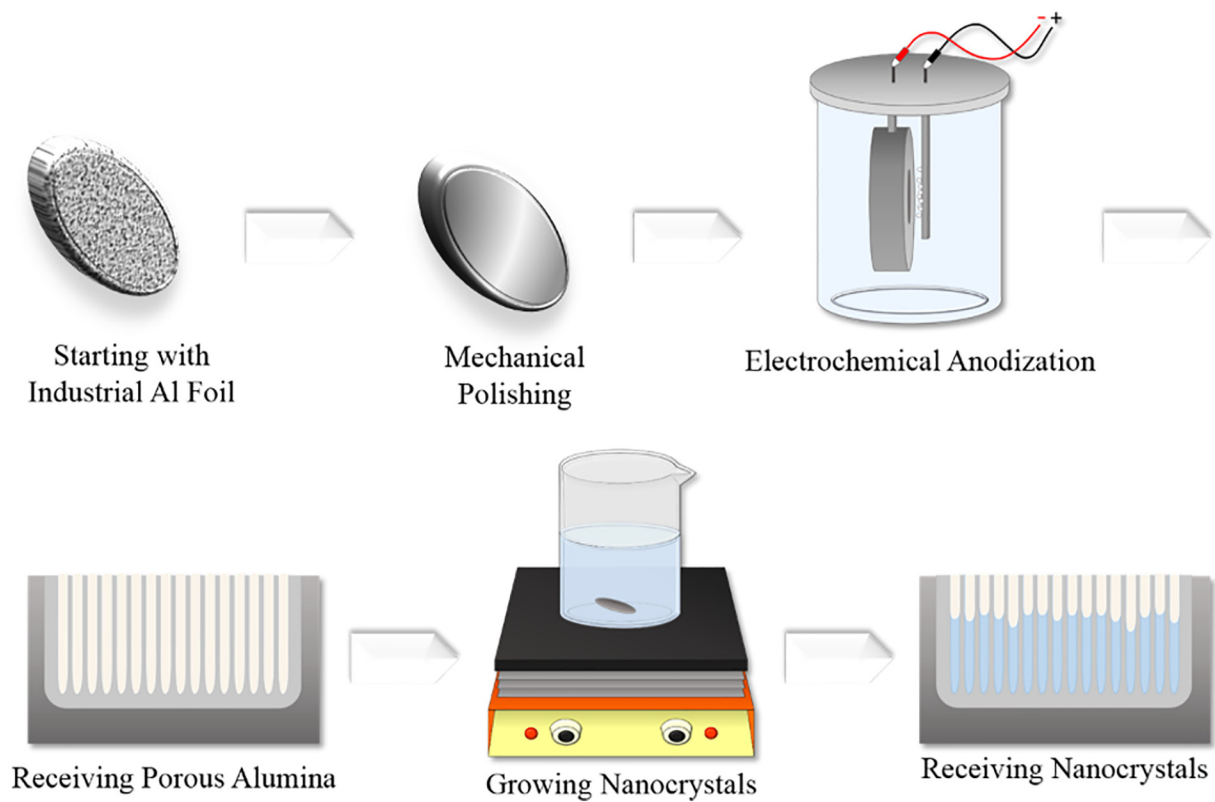


Fig. 1. A schematic illustration of the fabrication process.

Based on the XRD patterns (Fig. 4), only α -glycine crystals with pronounced preferred orientation of (010) planes parallel to the porous layer plane (ICDD card #00-032-1702 is used to identify the peak) are formed inside the pores at the highest liquid concentration of 2.3 g/10 g H_2O . A decrease of the liquid concentration to 2 g/10 g H_2O leads to the formation of β -glycine crystals along with α -polymorph crystals (the ICDD card #00-002-017 is used to identify the peak). A further decrease of the liquid concentration to 1.8 g/10 g H_2O , results in a pronounced decrease of the peak corresponding to the α -polymorph as compared to the β -polymorph peak (Fig. 4). The β -glycine nanocrystals have a

preferred crystallographic orientation, where their polar axis is aligned along the longitudinal axes of pores vertically to the free porous surface of the sample.

An overlap between the (020) peak of the β -polymorph and the (130) peak of the α -polymorph (Fig. 4) may present a doubt about the presence of the β -polymorph inside the pores. This is resolved by the piezoelectric current response (which will be presented in the piezoelectric measurements results further in this paper), which appears only in the β -polymorph crystals and not in the α -polymorph crystals.

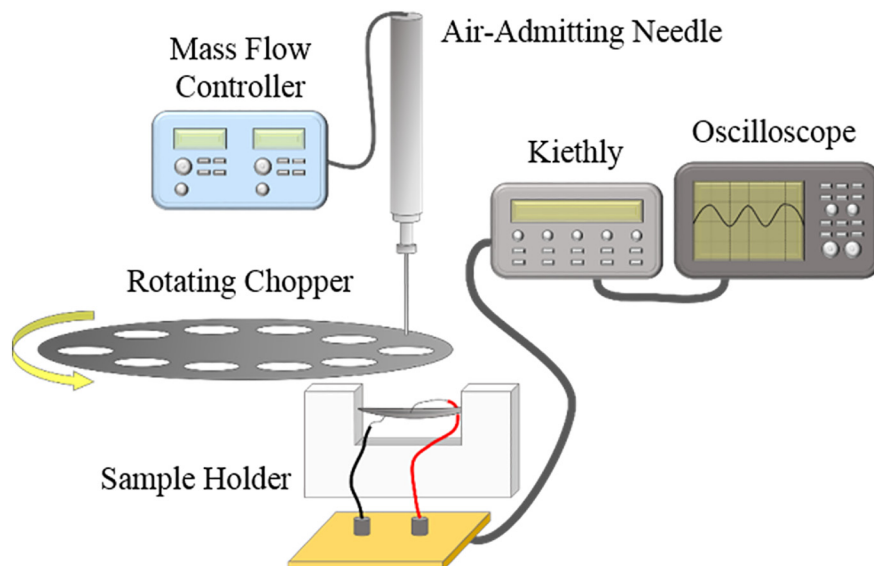


Fig. 2. A schematic illustration of the experimental setup used for measuring the piezoelectric current response. A mechanical pressure is applied by an air flow vertically directed towards the surface of the sample, its magnitude is controlled by a mass flow controller and modulated by a rotating wheel with holes.

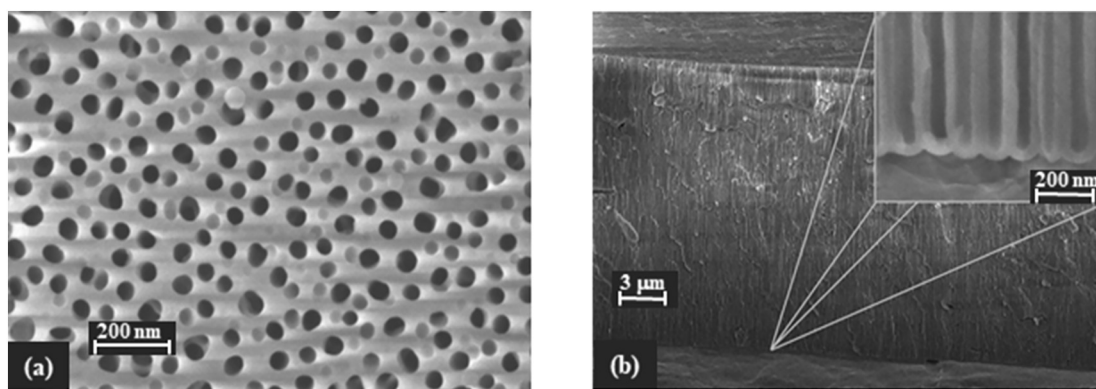


Fig. 3. HRSEM images of the porous layer in the aluminum substrate. (a) Top-view image and (b) vertical cross section view images.

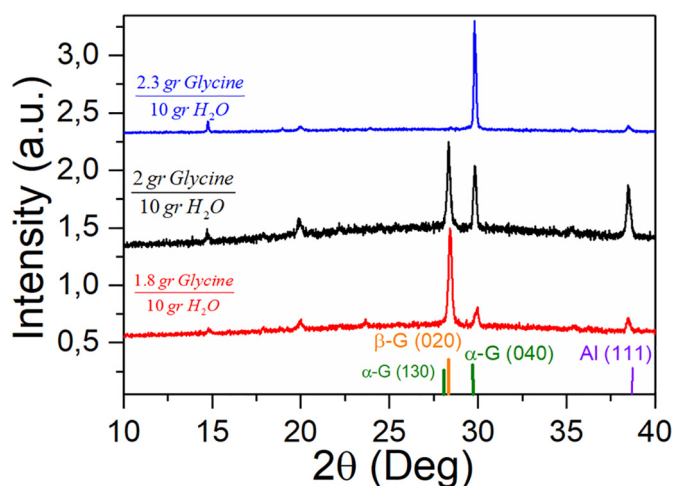


Fig. 4. XRD spectra of glycine crystals grown inside the alumina porous layer. The type of polymorph of glycine crystals and the corresponding crystallographic plane are indicated at each XRD peak along with the concentration of the liquid solution.

3.3. L-Alanine nanocrystals growth

Nanocrystals of L-alanine were grown inside the alumina pores with a preferred crystallographic orientation of (001) planes parallel to the free porous surface, based on the XRD patterns (Fig. 5).

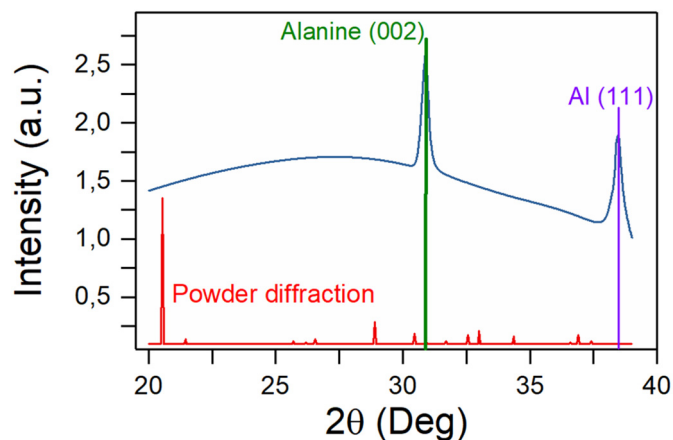


Fig. 5. XRD spectra of L-alanine crystals grown inside the alumina porous layer. The XRD diffraction peaks of randomly oriented L-alanine crystals in a powder and the Al substrate are also presented.

3.4. Piezoelectric current response to weak applied mechanical pressures

In order to ensure that measured properties correspond to the amino acid crystals instead of substrate, piezoelectric measurements were performed on porous samples where the alumina pores are not filled with the crystals with the same set up of electrical contacts and instrument. The empty AAO samples do not show any electric current response change to the alternating applied force. This result leads to a conclusion that the any electric response of a sample is associated with the crystals embedded inside the nano-pores.

The α-glycine crystals, formed inside the pores at the highest concentration of the liquid solution, do not show any electric current response to the alternating applied mechanical pressures. This is an expected result due to the centro-symmetric structure of the α-polymorph [25]. However, when the β-polymorph crystals are formed inside the pores at the liquid concentrations of 2 and 1.8 g/10 g H₂O, electric current response is measured. This result is expected due to the non-centro-symmetric unit cell of the β-polymorph crystals. The L-alanine crystals grown inside the pores show electrical response to the applied mechanical pressures, their time evolution of electrical current in response to the periodic applied pressure is shown in Fig. 6. Similar profiles are obtained for β-glycine and co-existence of α- and β-glycine, see Figs. S1 and S2 in Supplementary information. The lowest measurable signal by β-glycine and L-alanine are obtained at mechanical pressures of 1 and 3 Pa, respectively. Any applied pressure below the aforementioned values could not activate a measurable electrical signal by the crystals, hence electrical current is defined as zero in this case — within the limit 0.01 pA. No changes in the magnitude or shape of the current were observed in response to mechanical pressures applied for two weeks.

The amplitude of the electrical current response increases linearly when increasing the amplitude of the applied mechanical pressure, as shown in Fig. 7. This linearity confirms the fact that the measured electrical current is indeed piezoelectric current, taking into consideration the non-centro-symmetric structure of the materials. The dependence between the applied mechanical pressure and the piezoelectric current response is weaker for the L-alanine nanocrystals, compared to the β-glycine nanocrystals. The piezoelectric response of the latter phase is in accordance with former report by the authors [45].

4. Discussion

In this section, we discuss the crystal growth process of the materials, their piezoelectric response to weak applied mechanical pressures and potential applications. First, we explain the crystals growth process with preferred crystallographic orientation inside the alumina nanopores. Afterwards, the piezoelectric response of each material is analyzed based on the sample's geometry, crystallographic orientation and molecular structures of the crystals. Finally, the potential applications

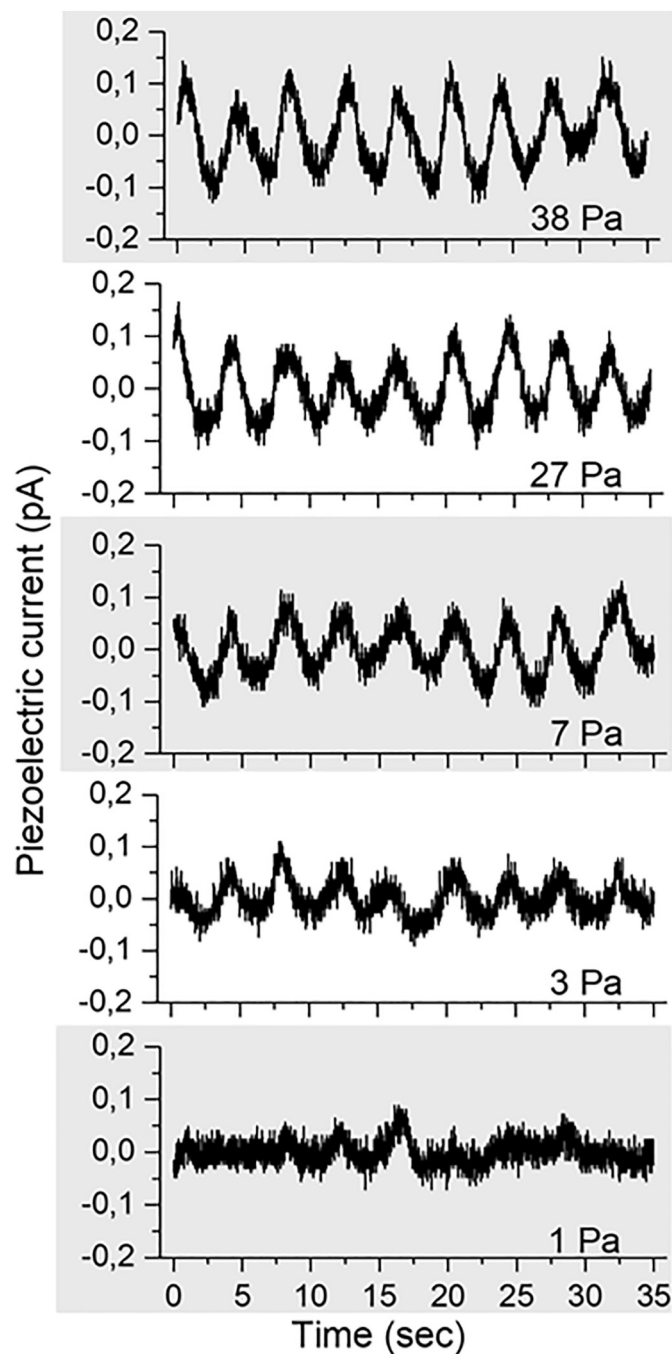


Fig. 6. The piezoelectric current response to time modulated applied mechanical pressures of L-alanine crystals grown inside the pores.

are presented.

Nanocrystals of glycine and alanine precipitate only inside the nanopores but not within the bulk liquid. This is due to a surface curvature effect, causing an enhanced solute concentration in the liquid solution within the nano-pores. The curved surface close to the bottom of the nano-pores causes a change in the thermodynamic state of the liquid solution in contact with this surface as compared to a straight surface outside the pores [55,56]; hence an increase of the solute concentration in the liquid solution in contact with this curved surface at the bottom of the pores occurs leading to oversaturation and precipitation of solute ions. This phenomenon of crystals nucleation and growth inside pores was observed in other inorganic crystals and reported in detail in previous papers [14,45,50,51].

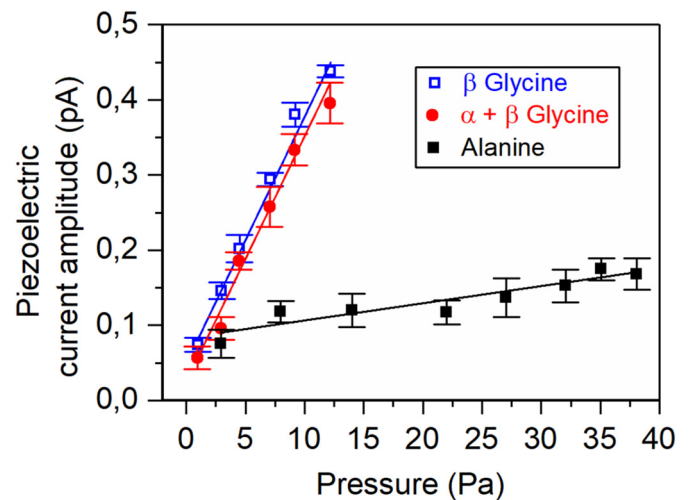


Fig. 7. The piezoelectric current response of β -glycine and L-alanine nanocrystals inside the alumina pores to alternating applied mechanical pressures.

Both glycine and alanine crystals grow inside the alumina pores with a pronounced preferred crystallographic orientation of [010] and [001] directions, respectively, along the longitudinal axes of the pores. These crystallographic directions are claimed to be determined by a kinetic preference, since the same preferred crystallographic orientations are identified in the bulk size crystals of β -glycine and L-alanine as a fast crystal growth directions [41,57]. The formation of β -glycine polymorph is attributed to a relatively low solute concentrations in the liquid solution, which is in accordance with previous literature reports on the crystals formation and growth [45,58], where the type of polymorph that nucleates and grows is determined by a competition between kinetic and thermodynamic driving forces [42,43,45,59].

The high detection sensitivity of ultra-low applied mechanical pressures of both β -glycine and L-alanine is attributed to their molecular structures, which involve hydrogen bonds between the organic molecules. Hydrogen bonds are easily deformed elastically when subjected to external mechanical pressures as compared to the covalent bonds within the molecules. The deformation of hydrogen bonds alters the dielectric polarization inside the materials unit cells, leading to a piezoelectric response. Such weak hydrogen bonds make the materials highly sensitive to ultra-low applied mechanical pressures.

Since the crystals of alanine and glycine crystals are uniformly oriented, their piezoelectric current response corresponds to the change in the dielectric polarization in the [010] and [001] crystallographic directions of the glycine and alanine crystals, respectively, along the longitudinal axis of the pores (e.g. Fig. 8). Therefore, measured electrical current corresponds to the polarization change in the above-mentioned directions. By taking into consideration the crystallographic orientations of the crystals inside the pores and their piezoelectric coefficients tensors (Eqs. (2) and (3)), the change in their dielectric polarization can be expressed as P_g in glycine (Eq. (4)) and P_a in alanine (Eq. (5)).

$$P_g = P_2 = d_{2j} \cdot \epsilon_j = d_{21} \cdot \epsilon_1 + d_{22} \cdot \epsilon_2 + d_{23} \cdot \epsilon_3 + d_{25} \cdot \epsilon_5 \quad (4)$$

$$P_a = P_3 = d_{3j} \cdot \epsilon_j = d_{36} \cdot \epsilon_6 \quad (5)$$

Eq. (5) implies that in the case of L-alanine, the piezoelectric effect is activated only by a shear strain within the plane parallel to the free surface of the sample. In the case of the β -glycine, the piezoelectric current is generated by both tensile or compressive and shear strains, as can be deduced from Eq. (4).

The fact that the piezoelectric response of each of the samples was conserved for two weeks shows that β -glycine polymorph is stabilized inside nano-pores. Meta-stability of β -phase inside pores has been

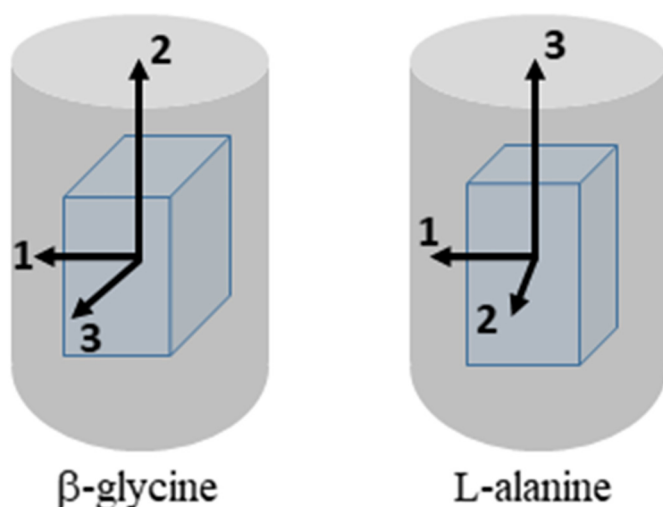


Fig. 8. Crystallographic orientations of β -glycine and L-alanine crystals in respect to the longitudinal axes of pores. Electrical measurements are performed along the cylindrical axes of the pores.

previously reported in literature. This effect is attributed to the confined size of crystals, as well as to the role of pores in protection of crystals from humidity, which enhances the transformation to a stable phase [60,61].

Highly sensitive piezoelectric response of amino acids crystals can be applied in bio-compatible mechanical pressure sensors in versatile types of medical devices implanted inside the body of humans or other living organisms. Such sensors may be used for air pressure monitoring for respiratory diseases, intra-uterine monitoring for obstetrics, intracranial pressure monitoring during neurosurgery, and pressure sensors for artificial implants [62,63]. The main advantage of using these bio-compatible piezoelectric sensors is their passive operation without a need for electrical batteries.

5. Summary

Detection of ultra-low mechanical pressures in the scale of 1 Pa by β -glycine and L-alanine nanocrystals grown inside alumina pores by a piezoelectric current response is confirmed experimentally. The nanocrystals were grown inside alumina nano-pores by precipitation from aqueous solutions. The crystals have a preferred crystallographic orientation along the longitudinal axis of the pores characteristic to the fastest growth rate direction. Controlling the crystal growth of glycine is highly important for obtaining the β -polymorph that has piezoelectric characteristics. This paper shows that β -glycine and L-alanine nanocrystals can be applied as bio-compatible mechanical pressure sensors.

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.msec.2019.110468>.

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